

BASIC DATA EXPLORATION AND ANALYSIS IN MICROSOFT EXCEL

Introduction

This dataset is a product sales and inventory sheet that contains information about different products, their brands, prices, quantities, and categories. The purpose of the analysis is to understand product distribution, pricing patterns, inventory levels, and apply Excel functions for business insights.

The dataset includes:

Product ID

Product Name

Brand Name

Price (\$)

Quantity

Category

The analysis uses Excel formulas such as:

SUM

COUNT

AVERAGE

MIN

MAX

IF

SUMIF

COUNTIF

LEFT / MID / RIGHT

Data Analysis

1. Total Price Analysis

To calculate the total value of all product prices:

SUM

Excel Formula

```
=SUM(D2:D35)
```

Result: Total Price = ₹10,100

Interpretation: The combined price value of all listed products is ₹10,100, showing the overall pricing size of the inventory.

2. Counting Total Products

COUNT

Excel Formula

```
=COUNTA(B2:B35)
```

Result

Total Products = 34

Interpretation: There are 34 products available in the dataset.

3. Average Product Price

AVERAGE

Excel Formula

```
=AVERAGE(D2:D35)
```

Result

Average Price ≈ ₹297

Interpretation: On average, products are priced around ₹297, indicating a mix of low-cost and high-cost items.

4. Minimum and Maximum Price

Minimum Price

Excel Formula

```
=MIN(D2:D35)
```

Result:

Minimum Price = ₹30

Maximum Price

Excel Formula

=MAX(D2:D35)

Result:

Maximum Price = ₹1000

Interpretation: The dataset contains both budget and premium products. The price range is wide, showing product diversity.

5. Logical Function (IF)

Excel Formula

=IF(D2:D35>500,"HIGH PRICE","STANDARD")

The screenshot shows an Excel spreadsheet titled 'assignment sheet - Excel'. The formula bar displays the formula **=IF(D2:D35>500,"HIGH PRICE","STANDARD PRICE")**. The spreadsheet contains two tables. The first table has columns: Product ID, Product Name, Brand Name, Price (\$), Category, and Price Range. The second table has columns: Product ID, DAY, MONTH, and COUNTRY CODE. The Price Range column in the first table is populated with 'HIGH PRICE' for products with a price greater than 500 and 'STANDARD PRICE' for others.

Product ID	Product Name	Brand Name	Price (\$)	Category	Price Range
28-JAN-US	Laptop	Dell	1000	Electronics	HIGH PRICE
15-FEB-US	Sneakers	Nike	80	Fashion	STANDARD PRICE
03-MAR-US	Coffee Maker	Keurig	130	Kitchen	STANDARD PRICE
11-APR-US	Smartphone	Samsung	900	Electronics	HIGH PRICE
22-MAY-US	Backpack	North Face	70	Outdoor	STANDARD PRICE
07-JUN-UK	Headphones	Sony	200	Electronics	STANDARD PRICE
19-JUL-UK	T-shirt	Adidas	30	Fashion	STANDARD PRICE
23-AUG-UK	Blender	Ninja	90	Kitchen	STANDARD PRICE
05-SEP-UK	Tablet	Apple	500	Electronics	HIGH PRICE
14-OCT-UK	Hiking Boots	Timberland	130	Outdoor	STANDARD PRICE
17-JUN-IN	Laptop	HP	950	Electronics	HIGH PRICE
25-NOV-AU	Sneakers	Adidas	90	Fashion	STANDARD PRICE
08-DEC-DE	Coffee Maker	Nespresso	120	Kitchen	STANDARD PRICE
18-FEB-CA	Smartwatch	Fitbit	150	Electronics	STANDARD PRICE
16-APR-ES	Headphones	Bose	250	Electronics	STANDARD PRICE
21-AUG-CA	Laptop Bag	Samsonite	50	Accessories	STANDARD PRICE
20-AUG-CN	Smartwatch	Huawei	160	Electronics	STANDARD PRICE
27-JAN-IT	Laptop	Asus	980	Electronics	HIGH PRICE
01-MAR-UK	Sunglasses	Oakley	150	Fashion	STANDARD PRICE

Interpretation

Products costing above ₹500 are classified as “HIGH PRICE,” while others are marked “STANDARD.” This helps identify premium products quickly.

6. Conditional Functions

SUMIF

Excel Formula

=SUMIF(F2:F35,"Electronics",D2:D35)

Result

Electronics Total Price = ₹8050

Interpretation: Electronics contribute the highest share of total pricing value in the dataset.

COUNTIF

Excel Formula

=COUNTIF(D2:D35,"<100")

Result

11 products are priced below ₹100

Interpretation : A significant number of products are low-cost items.

7. Text Formatting Functions

LEFT Function

Excel Formula

=LEFT(A2,2)

MID Function

Excel Formula

=MID(A2,4,3)

RIGHT Function

Excel Formula

=RIGHT(A2,2)

The screenshot shows an Excel spreadsheet with two tables. The first table, in columns A-F, lists products with details like ID, Name, Brand, Price, Quantity, Category, and Price Range. The second table, in columns J-M, breaks down dates into Day, Month, and Country Code. The formula bar at the top shows '=LEFT(I2,2)'.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range
28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE
15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARD PRICE
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARD PRICE
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARD PRICE
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARD PRICE
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARD PRICE
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARD PRICE
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARD PRICE
17-JUN-IN	Laptop	HP	850	25	Electronics	HIGH PRICE
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARD PRICE
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARD PRICE
18-FEB-CA	Smartwatch	Fibit	150	15	Electronics	STANDARD PRICE
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARD PRICE
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARD PRICE
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	STANDARD PRICE
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	STANDARD PRICE

Product ID	DAY	MONTH	COUNTRY CODE
28-JAN-US	28	JAN	US
15-FEB-US	15	FEB	US
03-MAR-US	03	MAR	US
11-APR-US	11	APR	US
22-MAY-US	22	MAY	US
07-JUN-UK	07	JUN	UK
19-JUL-UK	19	JUL	UK
23-AUG-UK	23	AUG	UK
05-SEP-UK	05	SEP	UK
14-OCT-UK	14	OCT	UK
17-JUN-IN	17	JUN	IN
25-NOV-AU	25	NOV	AU
08-DEC-DE	08	DEC	DE
18-FEB-CA	18	FEB	CA
16-APR-ES	16	APR	ES
21-AUG-CA	21	AUG	CA
20-AUG-CN	20	AUG	CN
27-JAN-IT	27	JAN	IT
01-MAR-UK	01	MAR	UK

assignment sheet - Excel

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General

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DATA DATAS NEW COLUMN

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Product ID	DAY	MONTH	COUNTRY CODE
28-JAN-US	Laptop	Dell	1000	30	Electronics	HIGH PRICE	28-JAN-US	28	JAN	US
15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARD PRICE	15-FEB-US	15	FEB	US
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARD PRICE	03-MAR-US	03	MAR	US
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11-APR-US	11	APR	US
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARD PRICE	22-MAY-US	22	MAY	US
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARD PRICE	07-JUN-UK	07	JUN	UK
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARD PRICE	19-JUL-UK	19	JUL	UK
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARD PRICE	23-AUG-UK	23	AUG	UK
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05-SEP-UK	05	SEP	UK
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARD PRICE	14-OCT-UK	14	OCT	UK
17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17-JUN-IN	17	JUN	IN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARD PRICE	25-NOV-AU	25	NOV	AU
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARD PRICE	08-DEC-DE	08	DEC	DE
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARD PRICE	18-FEB-CA	18	FEB	CA
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARD PRICE	16-APR-ES	16	APR	ES
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARD PRICE	21-AUG-CA	21	AUG	CA
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARD PRICE	20-AUG-CN	20	AUG	CN
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27-JAN-IT	27	JAN	IT
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	STANDARD PRICE	01-MAR-UK	01	MAR	UK

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15-FEB-US	Sneakers	Nike	80	15	Fashion	STANDARD PRICE	15-FEB-US	15	FEB	US
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	STANDARD PRICE	03-MAR-US	03	MAR	US
11-APR-US	Smartphone	Samsung	900	25	Electronics	HIGH PRICE	11-APR-US	11	APR	US
22-MAY-US	Backpack	North Face	70	20	Outdoor	STANDARD PRICE	22-MAY-US	22	MAY	US
07-JUN-UK	Headphones	Sony	200	45	Electronics	STANDARD PRICE	07-JUN-UK	07	JUN	UK
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	STANDARD PRICE	19-JUL-UK	19	JUL	UK
23-AUG-UK	Blender	Ninja	90	35	Kitchen	STANDARD PRICE	23-AUG-UK	23	AUG	UK
05-SEP-UK	Tablet	Apple	500	50	Electronics	HIGH PRICE	05-SEP-UK	05	SEP	UK
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	STANDARD PRICE	14-OCT-UK	14	OCT	UK
17-JUN-IN	Laptop	HP	950	25	Electronics	HIGH PRICE	17-JUN-IN	17	JUN	IN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	STANDARD PRICE	25-NOV-AU	25	NOV	AU
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	STANDARD PRICE	08-DEC-DE	08	DEC	DE
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	STANDARD PRICE	18-FEB-CA	18	FEB	CA
16-APR-ES	Headphones	Bose	250	20	Electronics	STANDARD PRICE	16-APR-ES	16	APR	ES
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	STANDARD PRICE	21-AUG-CA	21	AUG	CA
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	STANDARD PRICE	20-AUG-CN	20	AUG	CN
27-JAN-IT	Laptop	Asus	980	10	Electronics	HIGH PRICE	27-JAN-IT	27	JAN	IT
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	STANDARD PRICE	01-MAR-UK	01	MAR	UK

Interpretation:These functions help separate country codes, months, or serial formats from Product IDs for better categorization.

Findings

Electronics products dominate the dataset in value.

The highest-priced product costs ₹1000.

The lowest-priced product costs ₹30.

Average pricing indicates moderate product pricing overall.

Several products fall under budget pricing (<₹100).

Text functions improve data organization and extraction.

Conclusion

The dataset contains products from different categories and brands.

Excel functions helped analyze prices, quantities, and product details easily.

Overall, the analysis improves understanding of inventory and business performance
e.Overall, the dataset shows a strong concentration of electronics products and a balanced mix of premium and budget items.